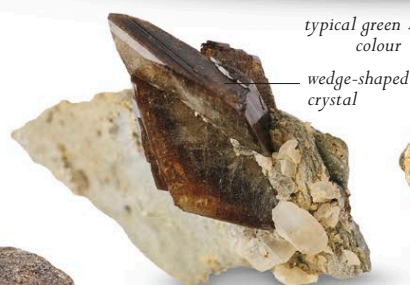




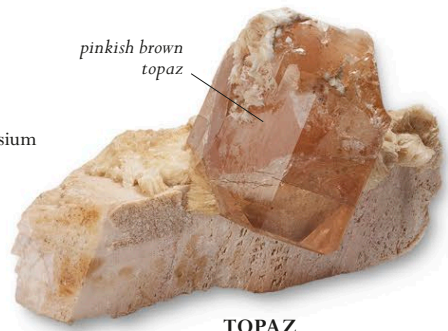
OLIVINE
Orthorhombic
H 7 • SG 3.27–4.39

Common in igneous rocks, nesosilicate minerals varying in composition from magnesium silicate to iron silicate are called olivine.

translucent



typical green colour
wedge-shaped crystal



pinkish brown topaz

TOPAZ
Orthorhombic • H 8 • SG 3.4–3.6

Topaz is an aluminium silicate fluoride hydroxide. Crystal size is usually small, but a giant crystal weighing 271 kg (596 lb) is known from Brazil.



CHLORITOID

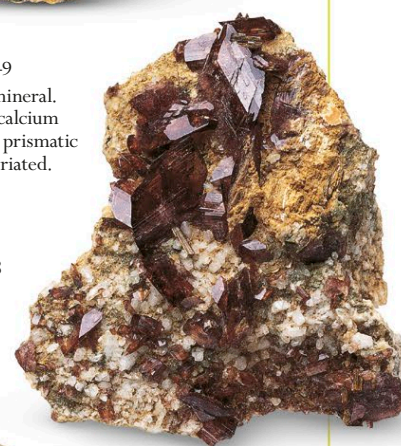
Monoclinic • H 6½ • SG 3.4–3.8
Widespread in metamorphic and volcanic rocks, chloritoid is a dark green or black hydrous aluminium silicate of iron, magnesium, and manganese.

SOROSILICATES



EPIDOTE

Monoclinic
H 6 • SG 3.38–3.49
Epidote is an abundant mineral. Crystals of this hydrous calcium aluminium iron silicate are prismatic or tabular, green, and striated.



AXINITE

Triclinic
H 6½–7 • SG 3.25–3.28
Axinite is a hydrous calcium iron manganese aluminium boron silicate, with axe-head-shaped crystals.

TWINNED
TITANITE
CRYSTALS

TITANITE

Monoclinic • H 5–5½ • SG 3.48–3.6
Variable in colour, titanite is a calcium titanium silicate. It is excellent at dispersing light – even better than diamond.

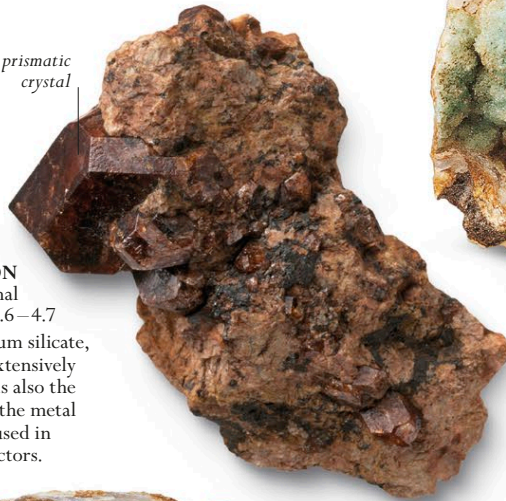
CRYSTALS
IN MATRIX



ANDALUSITE

Orthorhombic
H 6½–7½ • SG 3.13–3.21
Andalusite is an aluminium silicate. It occurs mainly in low-grade metamorphic rocks as coarse prismatic crystals with a square cross-section.

prismatic crystal



ZIRCON

Tetragonal
H 7½ • SG 4.6–4.7
Zircon, a zirconium silicate, is a gem used extensively in jewellery. It is also the main source of the metal zirconium, used in nuclear reactors.

short, prismatic
willemite crystal



WILLEMITE

Trigonal
H 5½ • SG 3.89–4.19
Willemite is a zinc silicate. White, green, yellow, or reddish, and usually massive, it occurs in altered zinc deposits and metamorphosed limestone.



long, parallel,
fibrous crystals

SILLIMANITE

Orthorhombic • H 6½–7½ • SG 3.23–3.27
Sillimanite is an aluminium silicate with long, slender crystals. Its composition is identical to that of andalusite, but it forms at higher temperatures and pressures.



HEMIMORPHITE

Orthorhombic
H 4½–5 • SG 3.48
This hydrated silicate of zinc occurs in altered zinc deposits. It is very variable in both its colour and form.

rounded
aggregates



DANBURITE

Orthorhombic
H 7–7½ • SG 2.93–3.02
Danburite is a calcium boron silicate. Its variably coloured crystals resemble those of topaz, but it can also be granular.



VESUVIANITE

Tetragonal
H 6½ • SG 3.32–3.43
Vesuvianite, also known as idocrase, is a hydrous calcium sodium magnesium iron aluminium silicate, with fluorine. It occurs in marble and igneous rocks.